

Pocket Sense: A Budget Tracker App

Problem Statement:

According to a study by Standard & Poor's (S&P Global Ratings), only around 25% of Filipino adults are considered financially literate. This means that many people in the Philippines are still prone to financial problems such as debt traps and poor cash flow management due to bad spending and budgeting habits. Many people lack a centralized, user-friendly system that can bridge the gap between financial intent and financial action. Without a tool that combines debt management, expense tracking, and budgeting, people will be still more likely to fall into a cycle of financial mismanagement. Therefore, a system is needed that can help and assist people in performing proper budgeting, managing their debt, and tracking their cash flow.

Project Objectives:

- To provide an interface for users to categorize expenses and set allocation limits
- To facilitate the daily annotation of expenses
- To assist in the annotation and management of debt

Planned Features:

- **Debt Tracker**
 - It tracks how much people owe to you and how much you owe to people. (**Ex:** Debt to Xanthus - ₱100)
 - It also takes into account the conditions and key details of the debt which is to be inputted by the user like the interest rate, due date/schedule for payment/installment, and the name of the person you owe/owes you. (**Ex:** Debt to Xanthus - ₱100, Interest rate - 10% interest, Time Interval - Weekly, Deadline: October 27)
 - It also has a debt forecasting section that will primarily forecast what-if scenarios regarding debt payment or debt growth at a certain rate in a menu type (**Example Question:** How long will I be able to pay my ₱500 debt with a 5% interest weekly if I pay in installments of ₱100 per week?)
- **Budgeting and Spending Tracker**
 - It allows for the creation of a general budget that is the combination of the categorization and allocation for the different expenses such as food, transportation, etc. It is also to be inputted by the user. (**Ex:** Food and Drinks - ₱500, School Stuff - ₱400, Transportation - ₱100 General Budget - ₱1000)
 - It has a platform where the user can annotate the expenses that they incur on different aspects of their budget. (**Ex:** Food and Drinks - ₱500, Rice and Viand Meal (5) - ₱60 each, Nestea (5) - ₱32 each, Graham Bar (1) - ₱35 each, Total Spending: ₱495)
 - It tracks and determines whether or not the total spending that has been annotated/entered in the platform has exceeded or is still under the allocated budget threshold of a specific aspect of the budget or the entire budget as a whole. (**Ex:** Food and Drinks - ₱500, Total Spending: ₱495, Remaining Allocation: ₱5, Remaining Total Budget: ₱505)

References:

- Klapper, L., Lusardi, A., & van Oudheusden, P. (2015). Financial literacy around the world: Insights from the Standard & Poor's Ratings Services Global Financial Literacy Survey. Global Financial Literacy Excellence Center.
https://gflec.org/wp-content/uploads/2015/11/Finlit_paper_16_F2_singles.pdf

Planned Inputs and Outputs:

Input

- If A, it will take you to another menu for the Debt Tracker (1)
- If B, it will take you to another menu for the Spending Tracker (2)

1.1 → If A, it will take you to an input process for your debt (what you owe to people(1.1.1))

→ If B, it will take you to an input process for other people's debt (what people owe you(1.2.2))

1.1.1 → asks for name input for who you owe.

1.1.2 → asks for the amount you owe

1.1.3 → whichever letter in the menu picked will be noted into a variable to be displayed later.

1.1.4 → Displays a short menu regarding the kind of interest for input (A,B,C)

1.1.4.A → Asks for input on how much percent of interest and calculates the amount.

1.1.4.B → Asks for input on how much money per time interval

2.1 → Asks for input on how much money you currently have.

2.2 → Asks for general_budget

2.3,2.4 loops until user inputs '='

2.3 → Asks for sub-budgets/ items with a budget

2.4 → Asks for amount of budget each item has

Output

→ Starting Menu (A or B) for whether you want to focus on the Debt Tracker or the Spending Tracker.

MENU 1

A. Debt Tracker

B. Spending Tracker

1.1 → Debt Tracker Menu (A or B) if you want to put it to your debt (what you owe to people(1.1.1) or other people's debt (what people owe you(1.2.2))

Debt Tracker MENU

- A. track your debt
- B. track other people's debt

1.1.1 → Displays "name:"

1.1.2 → Displays "write how much you owe."

1.1.3 → Displays a short menu regarding the time interval of the interest

(Example)

Is this ___?:

- A. Daily
- B. Weekly
- C. Monthly
- D. Quarterly
- E. Semi-Annually
- F. Annually
- G. Other
- H. N/A

1.1.3.A → variable 'time_interval' will have string amount 'Daily'

1.1.3.B → variable 'time_interval' will have string amount 'Weekly'

1.1.3.C → variable 'time_interval' will have string amount 'Monthly'

1.1.3.D → variable 'time_interval' will have string amount 'Quarterly'

1.1.3.E → variable 'time_interval' will have string amount 'Semi Annually'

1.1.3.F → variable 'time_interval' will have string amount 'Annually'

1.1.3.G → variable 'time_interval' will have string amount 'Other'

1.1.3.H → variable 'time_interval' will have string amount 'N/A'

1.1.4 → Displays a short menu regarding the kind of interest: (Is this ___?:)

- A. Percent (Ex: 5%)
- B. Other (Ex: +P20 per day past the deadline)
- C. Varies per interval
- D. N/A

1.1.4.A → Displays "how much percent?"

1.1.4.B → Displays "how much", time_interval, "?"

1.1.5 → Displays final output in this format: name, ":", how_much_you_owe, ":", interest_amount, time_interval

2.1 → Displays "Input how much money you currently have."
2.2 → Displays "What's your general budget?"
2.3, 2.4 *loops until the user inputs '='*
2.3 → Displays "Input budget item name."
2.4 → Display "Input budget amount."
2.4.1 → If the total expense is above the allocation or the general budget, display "You have overspent your budget by ₱____."
2.4.2 → If the total expense is under the allocation or the general budget, display "You have ₱____ remaining in your budget."
2.5 → Displays total amount spent

Logic Plan:

START

DISPLAY "MAIN MENU"

DISPLAY "A. Debt Tracker"

DISPLAY "B. Spending Tracker"

INPUT choice_main

IF choice_main == 'A' THEN

 // Debt Tracker Menu

 DISPLAY "DEBT TRACKER MENU"

 DISPLAY "A. Track your debt"

 DISPLAY "B. Track other people's debt"

 INPUT choice_debt

 IF choice_debt == 'A' THEN

 // Track your debt

 LOOP

 DISPLAY "Enter the name of the person you owe:"

 INPUT debtor_name

 DISPLAY "Enter the amount you owe:"

 INPUT amount_owed

 DISPLAY "Select time interval for interest:"

 DISPLAY "A. Daily"

 DISPLAY "B. Weekly"

 DISPLAY "C. Monthly"

 DISPLAY "D. Quarterly"

 DISPLAY "E. Semi-Annually"

 DISPLAY "F. Annually"

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DISPLAY "G. Other"
DISPLAY "H. N/A"
INPUT interval_choice
```

```
SWITCH interval_choice:
  CASE 'A': time_interval = "Daily"
  CASE 'B': time_interval = "Weekly"
  CASE 'C': time_interval = "Monthly"
  CASE 'D': time_interval = "Quarterly"
  CASE 'E': time_interval = "Semi-Annually"
  CASE 'F': time_interval = "Annually"
  CASE 'G': time_interval = "Other"
  CASE 'H': time_interval = "N/A"
```

```
DISPLAY "Select interest type:"
DISPLAY "A. Percent"
DISPLAY "B. Fixed per interval"
DISPLAY "C. Varies per interval"
DISPLAY "D. N/A"
INPUT interest_type
```

```
IF interest_type == 'A' THEN
  DISPLAY "Enter interest percent:"
  INPUT interest_percent
  interest_amount = amount_owed * (interest_percent / 100)
```

```
ELSE IF interest_type == 'B' THEN
  DISPLAY "Enter additional amount per", time_interval, ":"
  INPUT fixed_interest
  interest_amount = fixed_interest
```

```
ELSE IF interest_type == 'C' THEN
  interest_amount = 0
```

```
END IF
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```
DISPLAY debtor_name, ":", ₱, amount_owed, ", Interest: ₱", interest_amount, " per ",
time_interval
```

```
DISPLAY "Would you like to continue? [Y/N]"
INPUT continuing_choice
```

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    IF continuing_choice == 'N' THEN
        BREAK LOOP

ELSE IF choice_debt == 'B' THEN
    // Track others' debt to you
    DISPLAY "Enter the name of the person who owes you:"
    INPUT creditor_name

    DISPLAY "Enter the amount they owe you:"
    INPUT amount_owed_to_you

    // Optional: could reuse similar interest structure here

    DISPLAY creditor_name, " owes you ₱", amount_owed_to_you

END IF

ELSE IF choice_main == 'B' THEN
    // Spending Tracker
    DISPLAY "Enter how much money you currently have:"
    INPUT current_money

    DISPLAY "Enter your general budget:"
    INPUT general_budget

    total_spent = 0
    budget_items = EMPTY_LIST

    LOOP
        DISPLAY "Input budget item name (or type '=' to finish):"
        INPUT item_name

        IF item_name == '=' THEN
            BREAK LOOP

        DISPLAY "Input budget amount for ", item_name, ":"
        INPUT item_amount

        ADD (item_name, item_amount) TO budget_items
        total_spent = total_spent + item_amount

    END LOOP

    IF total_spent > general_budget THEN

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```
overspent_amount = total_spent - general_budget  
DISPLAY "You have overspent your budget by ₱", overspent_amount
```

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ELSE
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    remaining_budget = general_budget - total_spent  
    DISPLAY "You have ₱", remaining_budget, " remaining in your budget."
```

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END IF
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DISPLAY "Total spent: ₱", total_spent
```

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END IF
```

```
END
```